A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

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Abstract

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of d_F in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

1 Preliminaries

Definition 1 (Ordered vector space with cone)**.** Let V be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair (V, V^+) is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

Definition 2 (Pointed cone)**.** A cone V^+ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element v with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

Definition 3 (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone V^+ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple (V, V^+, e) is called an *Archimedean order unit space* (AOU space).

Definition 4 (Face of a cone)**.** Let V^+ be a cone in V and $x \in V^+$. The *face of V^+ generated by x* is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{ w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of V^+ that can be "majorised" by a positive multiple of x. This is the smallest face of V^+ containing x in its relative interior.

Remark 5. In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone PSD_n, the face generated by a rank-r matrix X is isomorphic to PSD_r, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or r^2 (complex Hermitian case).

2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

Theorem 6 (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

If $X \in C$ is an extreme point of C, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.

Proof. Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every t.

Since $B \succ 0$ and D is a non-zero symmetric matrix with at least one negative eigenvalue (replacing D by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts X being an extreme point of C (since X is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen t_0'), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Pointedness** of the cone, to guarantee that a line through an interior point of a face must exit the cone;
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Theorem 7 (Barvinok–Pataki for pointed cones)**.** *Let V be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of V^+ generated by x. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

If x is an extreme point of C, then

$$\boxed{d_F \le m.}$$

Equivalently, $\dim(F) \le m - 1$ (where $\dim(F)$ is the dimension of F as a convex set, i.e. $\dim(\mathrm{aff}(F))$).

Proof. By rank–nullity applied to φ_F,

$$d_F \;=\; \dim(\ker\varphi_F) + \dim(\operatorname{im}\varphi_F) \;\le\; \dim(\ker\varphi_F) + m.$$

It suffices to show that $\ker\varphi_F = \{0\}$ when x is extreme in C. Suppose for contradiction that there exists $D \in \ker\varphi_F$, $D \neq 0$.

Step 1: The perturbation preserves feasibility. Since $D \in \operatorname{span}(F) \cap \ker\varphi_F$, we have $\alpha_j(D) = 0$ for all j. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in V^+.

Step 2: Pointedness forces exit from the cone. Since V^+ is pointed, we cannot have both $D \in V^+$ and $-D \in V^+$ (this would imply $D = 0$). Therefore, at least one of the half-lines $\{x + tD : t > 0\}$ or $\{x - tD : t > 0\}$ must eventually leave V^+.

More precisely: since x lies in the relative interior of F, for sufficiently small $|t|$ we have $x + tD \in F \subseteq V^+$ (as $D \in \operatorname{span}(F)$ and x is an interior point of F). But the line cannot remain in V^+ in both directions indefinitely, since this would imply D and $-D$ are both in V^+ (by rescaling), contradicting pointedness.

Step 3: Walking to a smaller face. Let

$$t^+ = \sup\{t > 0 : x + tD \in V^+\}, \qquad t^- = \sup\{t > 0 : x - tD \in V^+\}.$$

By Step 2, at least one of t^+, t^- is finite. Say $t^+ < \infty$ (the other case is symmetric).

If V^+ is closed (which holds in finite dimensions, or under the Archimedean assumption in the norm topology), then $x^* := x + t^+D \in V^+$, and x^* lies on the boundary of V^+ relative to $\operatorname{span}(F)$. This means x^* belongs to a *proper face* $F' \subsetneq F$, so $\dim(F') < \dim(F)$. Moreover, x^* satisfies all constraints, so $x^* \in C$.

Step 4: Contradiction with extremality. We have $x^* = x + t^+D \in C$ and (if $t^- > 0$, which holds since x is interior to F) also $x^{**} = x - \varepsilon D \in C$ for small $\varepsilon > 0$. Then

$$x = \frac{\varepsilon}{t^+ + \varepsilon}\, x^* + \frac{t^+}{t^+ + \varepsilon}\, x^{**},$$

exhibiting x as a non-trivial convex combination of two distinct elements of C. If x were an extreme point of C, this is a contradiction.

Therefore $\ker\varphi_F = \{0\}$, and we conclude $d_F \le m$. □

4 Application to C*-Algebras and AOU Spaces

Corollary 8 (Barvinok–Pataki for AOU spaces)**.** *Let (V, V^+, e) be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \big\{\, x \in V^+ \;\big|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\big\}.$$

For any extreme point x of C, the face of V^+ generated by x satisfies $\dim(\operatorname{face}_{V^+}(x)) \le m - 1$.

Proof. The Archimedean property implies the cone V^+ is closed (in the order-unit norm) and pointed. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

Corollary 9 (Barvinok–Pataki for C*-algebras)**.** *Let A be a unital C^*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \big\{ \varphi \in S(A) \;\big|\; \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \big\},$$

where $S(A) = \{\varphi \in A^* : \varphi \geq 0,\ \varphi(\mathbf{1}) = 1\}$. If φ is an extreme point of C, then

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \;\leq\; m + 1,$$

where π_φ is the GNS representation of φ. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \leq m + 1$.

Proof. The state space $S(A)$ is a base for the cone A_+^* (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on A_+^* is $m + 1$ (the m given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by φ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\big(\mathrm{face}_{S(A)}(\varphi)\big) = \dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} - 1.$$

Applying Theorem 7 with the $m+1$ constraints on A_+^* gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m+1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m + 1$. □

5 The Infinite-Dimensional Case and the Role of d_F

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

Theorem 10 (Weis–Shirokov [4, Theorem 3.3])**.** *Let V be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let C denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of C generated by x has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of K generated by x has dimension m or $m + 1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if C is the intersection of ℓ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension m, then $\dim(F_K(x)) \in \{m, m + 1, \ldots, m + \ell\}$.

Corollary 11 (Finiteness of d_F)**.** *Let K be a convex set in a real vector space V, and let C be defined by ℓ generalised affine constraints on K. If x is an extreme point of C, then*

$$\dim\big(F_K(x)\big) \;\leq\; \ell.$$

In particular, the face of K generated by any extreme point of C is finite-dimensional, regardless of $\dim(V)$.

Proof. Since x is extreme in C, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 10, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

Remark 12 (No operator-algebraic assumptions needed)**.** Theorem 10 and Corollary 11 require only a real vector space V and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that d_F is finite.

- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on d_F, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 11 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on d_F.

Corollary 13 (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let V be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If x is an extreme point of C, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

Proof. By Corollary 11 (with $K = V^+$ viewed as a convex set and $\ell = m$), d_F is finite. Theorem 7 then gives $d_F \leq m$. □

6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m+1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.
2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m+1$: extreme measures under m moment constraints are supported on at most $m+1$ points.
3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m+1$ simultaneously constrains ranks within blocks *and* the number of active blocks.
4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under m constraints (including possibly unbounded Hamiltonians, handled via the Weis–Shirokov framework): $r^2 \leq m+1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

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